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Skills

Languages TypeScript, Python	Frontend React, Next.js, Tailwind CSS	Backend Node.js, Express.js, REST APIs, Event-Driven Architecture, Async Processing
Databases PostgreSQL, MongoDB, Redis	Async & Messaging BullMQ, Dead-Letter Queues, Idempotent Processing	Infrastructure Docker, GitHub Actions CI/CD, Vercel, DigitalOcean

Education

Faculty of Pharmacy 2024 Modern University

Work Experience

Vogel Edits 2025-01 - Present
Full-Stack Engineer (Adobe Extensions)

- Built and shipped production Adobe After Effects and Premiere Pro extensions, including Vanish BG, a real-time AI background removal tool available at <https://aescrpts.com> with ML inference running inside a sandboxed CEP environment.
- Designed a Node.js sidecar process architecture to bypass CEP runtime limitations, handling AI inference outside the sandbox and streaming results back to a React UI in real time over local HTTP IPC.
- Built and submitted a Premiere Pro TTS extension using Node.js with local ONNX model inference, designing the audio pipeline from text input to a rendered timeline clip.
- Collaborated asynchronously with designers and stakeholders, shipping production-ready tools using agile workflows.

Freelance 2024-11 - 2025-01
Full-Stack Engineer

- Built a full-stack tourism booking platform using Next.js, TypeScript, and Node.js with TypeORM for type-safe PostgreSQL access.
- Applied PostgreSQL indexing strategies and EXPLAIN ANALYZE to achieve sub-100ms API response times under concurrent load.
- Integrated Paymob payment gateway with idempotent transaction handling and webhook verification, reducing checkout friction by 40%.
- Improved SEO and Core Web Vitals to 95+ Lighthouse score using Next.js SSR, image optimization, and lazy loading.

Projects

RootCluster — AI-Powered Log Analysis Platform <https://rootcluster.dev>
Tech Stack: Node.js, TypeScript, PostgreSQL (pgvector), Redis, BullMQ, ONNX, AWS ECS Fargate, Docker, GitHub Actions

- Designed and deployed a distributed log ingestion pipeline on AWS ECS Fargate processing 1M+ events/hour with sub-second latency.
- Reduced ONNX embedding time from 20+ minutes to under 1 minute via batch processing, Redis caching, and worker autoscaling.

- Designed an event-driven pipeline (normalize, embed, cluster, merge, root cause) with retries, exponential backoff, dead-letter queues, and idempotent processing.
- Implemented centroid-based pgvector clustering to unify cross-service failures into single root causes.
- Set up CI/CD with GitHub Actions OIDC, ECR, SSM Parameter Store, RDS PostgreSQL, and ElastiCache.
- Applied PostgreSQL internals (MVCC, isolation levels, indexing) to build a high-throughput write pipeline while avoiding lock contention.

Medical CRM — AI-Enhanced Prescription Platform

Tech Stack: React, Node.js, PostgreSQL, JWT, RBAC

- Built a full-stack CRM handling diagnoses, prescriptions, and patient records.
- Implemented JWT auth with access/refresh token rotation and RBAC across three permission levels.
- Designed drug interaction validation across 500+ medications using a graph-based PostgreSQL lookup.
- Optimized dashboard queries for 10K+ monthly records using composite indexes and window functions.